

Ride-Sharing Demand Prediction Using GPS and Trip Data

A real-time Big Data architecture for zone-level ride-demand forecasting, from streaming ingestion to a predictive heatmap.

Kafka · Spark Structured Streaming · H3 / Geohash · Cassandra / Redis · XGBoost / LSTM

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Problem Statement

Ride-sharing platforms continuously generate GPS pings, trip requests, cancellations, and driver-availability updates. Traditional, batch-oriented data systems cannot process this volume and velocity fast enough to reflect real, location-specific demand — causing drivers to miss emerging demand and riders to face longer wait times.

01

High Volume & Velocity

Thousands of GPS pings and trip events arrive every second from a live, city-wide fleet.

02

Geospatial Complexity

Demand differs street-by-street and zone-by-zone — not just city-wide averages.

03

Real-Time Requirement

Batch analytics runs too late to capture fast-shifting, minute-to-minute demand.

04

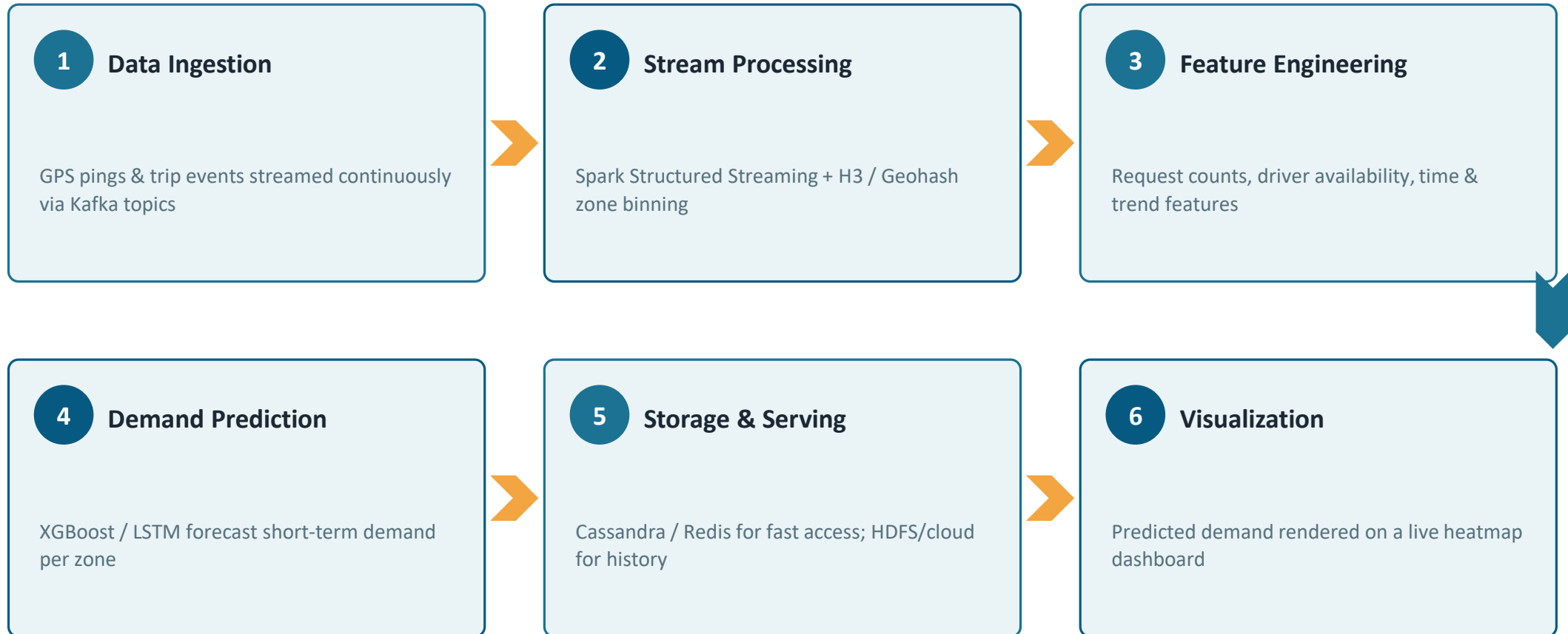
Supply–Demand Mismatch

Drivers rarely learn about emerging high-demand zones in time to reposition.

Goal: Design a scalable Big Data architecture that ingests, processes, and predicts ride demand at the zone level in near real time.

System Workflow

Raw GPS / trip events → streaming pipeline → zone-level demand prediction → heatmap dashboard



Note: historical trip data can be replayed through Kafka at accelerated speed to simulate a live GPS feed for demonstration.

Technology Stack

Existing, proven technologies — the novelty is in how they're integrated into one real-time architecture.

INGESTION LAYER

Apache Kafka

Real-time ingestion of GPS & trip events

PROCESSING LAYER

Spark Structured Streaming

Distributed streaming computation & aggregation

GEOSPATIAL INDEXING

H3 / Geohash

Divides the city into uniform geographic zones

HISTORICAL STORAGE

HDFS / Cloud Storage

Stores raw & processed data at scale

SERVING LAYER

Cassandra / Redis

Low-latency access to processed predictions

ML MODELS

XGBoost / LSTM

Short-term demand forecasting per zone

Visualization: predicted demand is rendered on a geographic heatmap dashboard (e.g., Plotly, Kepler.gl, or Streamlit).



Thank You

Questions & Discussion

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